

Evaluating Layer 2 Blockchain Scalability through a Comparative Study of Rollups and State Channels

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Abstract

Blockchain technology faces a persistent scalability limitation that restricts its adoption for large-scale decentralized applications. Layer 1 networks such as Bitcoin and Ethereum can process only a few transactions per second because every node must verify every block, creating latency and high transaction costs. This research evaluates the effectiveness of two major Layer 2 scaling approaches Rollups and State Channels in addressing this challenge. Using a systematic literature review of twenty-nine peer-reviewed studies, the paper analyses their design architectures, performance metrics, and security assumptions. Quantitative assessment is supported by mathematical models, including a throughput ratio

$$R = \frac{TPS_{L2}}{TPS_{L1}}, \quad \text{gas cost per transaction } C_{tx} = (G_{blob}/n) + G_{meta}$$

to measure relative scalability, and a simplified gas-cost model to estimate transaction efficiency. Findings indicate that Rollups can increase throughput by more than two orders of magnitude while preserving Layer 1 security, whereas State Channels achieve instantaneous confirmation for bilateral transactions but suffer from liquidity and routing constraints. The paper concludes that a hybrid architecture combining Rollups for global computation and Channels for micro-transactions provides the most balanced route toward Web-scale blockchain systems.

Keywords: Blockchain scalability, Layer 2, State channels, Rollups, Optimistic rollup, ZK-rollup, Throughput, Data availability, EIP-484

1 Introduction

The evolution of blockchain technology has transformed digital systems by enabling distributed trust without centralized intermediaries. However, as the number of users and applications has expanded, the fundamental design of early blockchains has exposed a critical limitation scalability. Public networks can sustain decentralization and security only by restricting throughput, which leaves them unable to match the capacity of conventional payment infrastructures. This trade-off is widely known as the *blockchain trilemma*, which states that decentralization, security, and scalability cannot all be maximized simultaneously (Zhou et al. 2020).

Current Layer 1 (L1) systems such as Bitcoin and Ethereum process approximately 7 to 30 transactions per second (TPS), far below

centralized processors that routinely exceed thousands of TPS. The limitation arises because each node must execute and validate every transaction to maintain consensus integrity. As a result, network congestion leads to unpredictable fees and long confirmation delays, undermining usability for global adoption (Rebello et al. 2024).

Researchers and developers therefore focus on Layer 2 (L2) frameworks that build atop existing blockchains to extend performance without sacrificing base-layer security. These frameworks handle the majority of operations off-chain and record only summary proofs or commitments on the main chain (Song et al. 2024). By transferring computational load away from the consensus layer, L2 protocols can scale throughput by several orders of magnitude while maintaining verifiable linkage to the base ledger.

Among L2 innovations, State Channels and Rollups represent two dominant paradigms. State Channels allow repeated off-chain exchanges between predefined participants, whereas Rollups aggregate thousands of transactions into compact batches that are periodically posted to L1. Both approaches share the objective of increasing throughput and reducing transaction cost, yet they differ sharply in architecture, trust model, and economic efficiency. To express scalability quantitatively, this study adopts the throughput ratio:

$$R = \frac{TPS_{L2}}{TPS_{L1}},$$

where TPS_{L2} and TPS_{L1} denote Layer 2 and Layer 1 transaction rates respectively. A higher R value indicates a more efficient scaling strategy.

Despite numerous surveys on blockchain scalability, a systematic comparative analysis of these two mechanisms evaluating their security proofs, data-availability models, and user-experience implications remains limited (Sanka and Cheung 2021). This paper addresses that gap through a critical synthesis of recent academic literature, performance metrics, and experimental outcomes. The following sections outline the methodology of literature selection, comparative evaluation of Rollups and State Channels, an in-depth discussion of their scalability trade-offs, and concluding recommendations for future Layer 2 development.

2 Methodology

The methodology provides a structured framework for assessing performance, security, and efficiency between rollups and state channels before detailing specific analytical steps in the following subsections.

2.1 Analytical Framework

According to Chaliasos et al. (2024a), effective comparative evaluation of blockchain scalability should rely on structured and measurable performance indicators rather than narrative summaries. Following this principle, the present study applies a comparative analytical framework to examine twenty-nine peer-reviewed publications on Layer 2 (L2) scalability. The analysis focuses on three

dimensions: quantitative performance, qualitative architecture, and contextual applicability. Quantitative metrics include transaction throughput, latency, and gas-cost efficiency, while qualitative factors assess trust models, data availability, and user experience.

The selected studies were grouped into three mechanism types: Optimistic Rollups, Zero-Knowledge (ZK) Rollups, and State Channels. Each category was evaluated using mathematical formulations common across literature, such as the throughput ratio and the transaction cost model. The formulas were used to standardize comparative analysis across multiple implementations reported in the reviewed studies.

2.2 Data Synthesis and Interpretation

As noted by Negka and Spathoulas (2021), State Channels differ from Rollups in their operational structure because they depend on user participation and liquidity availability. Building on that insight, each paper was coded according to its scalability dimension, verification model, and type of off-chain execution. Performance ranges were normalized to comparable values and summarized in analytical tables. When metrics conflict between sources, a triangulation process was applied, averaging results or confirming data consistency through cross-study validation.

The analysis also considered the context of each contribution, giving higher interpretive weight to works that included experimental data or formal proofs. This ensured that the overall synthesis captured both technical and empirical reliability rather than publication frequency alone.

2.3 Reliability and Bias Control

As highlighted by Tremblay Thibault et al. (2022), maintaining consistency across heterogeneous studies is essential for avoiding over-generalization in scalability research. To minimise bias, all extracted information was independently verified, and inclusion decisions were recorded for transparency. A sensitivity review was conducted to confirm that removing outlier studies did not materially change conclusions.

The final set of twenty-nine publications formed a robust evidence base for subsequent comparative evaluation in Sections 3 and 4.

This refined methodology ensures that findings are reproducible, balanced, and aligned with recognized academic standards for computing research.

3 Comparative Evaluation of Layer 2

Solutions

Layer 2 scalability mechanisms are designed to extend blockchain throughput without compromising decentralization or base-layer security. Among the numerous proposals, three approaches Optimistic Rollups, Zero-Knowledge (ZK) Rollups, and State Channels represent the most mature and widely implemented categories. Each employs a distinct method for shifting execution away from the congested Layer 1 chain. The following subsections analyze their architecture, performance behavior, and limitations.

3.1 Optimistic Rollups

Optimistic Rollups (ORs) scale blockchains by executing transactions off chain while posting compact batch data and a resulting state root to the base chain. The protocol *optimistically* assumes that submitted batches are valid; fraud is detected only when a participant challenges an incorrect batch within a fixed dispute window. Early architectural insight came from Arbitrum, which introduced an interactive verification protocol allowing the blockchain to replay only the disputed segment of execution rather than the entire transaction set (Kalodner et al. 2018). This approach drastically reduced on-chain computation and gas expenditure, enabling throughput improvements of one to two orders of magnitude.

However, the optimistic model introduces delayed finality. To guarantee security, users must wait through a challenge period typically about one week before withdrawals become irreversible. The long delay negatively affects user experience and liquidity. Quantified this drawback, showing that capital can remain locked for seven days while still achieving an average throughput exceeding 2 000 TPS, a significant advance over Layer 1's ≈ 30 TPS (Moosavi et al. 2023).

Metric	Typical Value
Throughput	≈ 2000 TPS
Withdrawal Delay	~ 1 week

Fee Reduction	10–100 $\times$ vs L1
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Table 1: Summary of ORs

Recent studies propose improvements to fraud-proof efficiency and early detection. The Early Fraud Detection (EFD) framework introduces lightweight probabilistic checks and local replay analysis to identify inconsistent state transitions before the full dispute period expires, reducing perceived latency (Gupta et al. 2024). Mathematically, scalability can be expressed by the ratio

$$R_{OR} = \frac{TPS_{OR}}{TPS_{L1}} \approx 100,$$

indicating hundred-fold throughput relative to the base chain. In summary, Optimistic Rollups offer near-complete EVM compatibility and high-cost efficiency but retain economic latency due to their reliance on post-factum verification.

3.2 Zero-Knowledge Rollups

Zero-Knowledge (ZK) Rollups differ fundamentally in verification strategy. Instead of assuming correctness, they employ succinct cryptographic proofs zk-SNARKs or zk-STARKs to demonstrate that every off-chain transaction batch has been executed faithfully. A valid proof guarantees state integrity without requiring disputes or waiting periods (Sun et al. 2021).

According to Chaliasos et al. (2024a) benchmarked contemporary zk-Rollup frameworks and confirmed that proof-generation cost, rather than on-chain gas, is the dominant performance constraint. While verification on Layer 1 takes only milliseconds, producing proof for thousands of transactions may require several seconds of heavy computation. Despite this overhead, ZK-Rollups achieve faster finality because validity is established *a priori*. Their security relies on the mathematical soundness of the underlying proof system, offering the strongest theoretical assurance among Layer 2 models.

Performance evaluations indicate that current zk-Rollups attain between 1 000 and 3 000 TPS with fees 80–90 % lower than base-layer gas costs once data compression and EIP-4844 “blob” transactions are used (Saif et al. 2024). A simplified expression for gas efficiency can be written as

$$C_{tx} = \frac{G_{data}}{N_{batch}} + G_{meta},$$

where G_{data} denotes gas for posting batch data, N_{batch} is the number of transactions, and G_{meta} the minimal verification overhead. Lower C_{tx} reflects greater cost reduction.

Metric	Typical Value
Throughput	1 000–3 000 TPS
Finality	< 1 minute
Cost Reduction	> 80 % post-EIP-4844

Table 2: Summary of ZK

Although trusted-setup ceremonies and high prover costs currently hinder universal deployment, continuous advances in hardware acceleration and recursive proofs are mitigating these barriers. ZK-Rollups therefore deliver immediate finality, cryptographic security, and superior cost efficiency, making them ideal for high-value or compliance-sensitive applications such as decentralized exchanges and cross-chain settlements.

3.3 State Channels

State Channels predate rollups and represent the earliest practical Layer 2 method for off-chain execution. They enable two or more participants to lock funds in a smart-contract escrow and exchange signed state updates privately. Only the opening and closing transactions are recorded on-chain, yielding near-instantaneous settlement for the participants (Negka and Spathoulas 2021). Because intermediate states are exchanged directly, channel throughput is limited only by network latency, achieving effective transaction rates exceeding 10 000 TPS between parties.

The Sprites framework enhanced earlier designs by eliminating incremental time-locks for multi-hop payments. Instead of increasing settlement delay linearly with hop count, Sprites maintains a constant lock-time across routes, thereby reducing both latency and collateral cost (Miller et al. 2019). This improvement allows large payment networks, such as Lightning or Raiden, to scale with minimal capital overhead. A

simple liquidity-efficiency measure can be expressed as

$$E_{cap} = \frac{V_{tx}}{L_{locked} \times t_{lock}},$$

where V_{tx} denotes total transaction volume, L_{locked} the collateral deposited, and t_{lock} the average lock duration. Higher E_{cap} indicates better use of capital.

Metric	Typical Value
TPS (per channel)	10 000 +
Latency	< 1 s
Fee per Tx	≈ 0 (on-chain open/close only)

Table 3: Summary of State Channels

Nevertheless, channels depend on *liveness*: every participant must remain online or delegate monitoring to an external service. The Pisa protocol introduced *watchtowers* that monitor the blockchain and automatically dispute fraudulent closures, protecting offline users (McCorry et al. 2019). While channels offer unmatched privacy and instant confirmation, their usability declines in large, open networks requiring frequent channel creation or liquidity rebalancing. Consequently, State Channels are optimal for repeated bilateral interactions such as micropayments or gaming where participants are known in advance, complementing rollup systems that serve broader public applications.

4 Problem Analysis and Discussion

This section interprets the comparative evidence by focusing on five critical dimensions: throughput & latency, security & trust, cost efficiency, scalability boundaries with hybrid architecture, and user experience and interoperability. Each subsection applies measurable indicators and synthesizes insights from the literature to evaluate practical scalability.

4.1 Throughput and Latency

Throughput and latency determine the operational viability of any Layer 2 solution. Rebello et al. (2024) report that contemporary Layer 2 systems can increase Ethereum’s transaction rate from roughly 15 TPS to

between 2 000 and 5 000 TPS, a throughput ratio

$$R = \frac{TPS_{L2}}{TPS_{L1}} \approx 100\text{--}300.$$

This metric captures the proportional efficiency of off-chain execution.

Empirical benchmarks compiled by Tremblay Thibault et al. (2022) show that ZK-Rollups generally offer lower confirmation latency (< 1 minute) because validity proofs are verified immediately on-chain, whereas Optimistic Rollups experience finality delays of about one week due to challenge windows. State Channels achieve near-zero latency for bilateral transactions but are bound by network routing delays when scaled to multi-hop payments.

Qi et al. (2024) emphasize that latency improvement is context-dependent: while channels provide instantaneous responses for small groups, rollups deliver consistent performance for thousands of concurrent users. Overall, Layer 2 mechanisms successfully transform blockchain throughput from a single-lane process into multi-lane architecture, trading occasional setup delay for persistent high-speed transaction flow.

4.2 Security and Trust Assumptions

Security in Layer 2 protocols arises from how faithfully off-chain computation can be verified and how resistant systems are to data withholding. Tortola et al. (2024) classify proof mechanisms into *fraud-proof* and *validity-proof* schemes: Optimistic Rollups depend on the former, requiring at least one honest verifier to detect errors, whereas ZK-Rollups rely on mathematically sound validity proofs that make incorrect states cryptographically impossible.

A persistent risk is data-availability (DA) the requirement that users can access transaction data to reconstruct state if an operator disappears. Huang et al. (2024) propose distributed storage using erasure-coded shards to guarantee DA and decentralization in zk-systems. Their method mitigates single-operator failure by dispersing data across independent validators.

Motepalli et al. (2023) address another trust component: sequencer centralization. Most rollups currently rely on a single sequencer to order transactions. Decentralized-sequencer frameworks rotate or randomize the sequencing

authority, reducing censorship risk and aligning economic incentives with network honesty.

The combined findings suggest that Layer 2 security mirrors a spectrum from human-dependent detection in Optimistic Rollups, to mathematically enforced validity in ZK-Rollups, to participant-monitored safety in State Channels. Each model trades computational complexity for differing trust assumptions, allowing designers to tailor assurance levels to application sensitivity.

4.3 Cost Efficiency

Transaction cost is a decisive factor for scalability adoption. Rollup gas expenditure is largely governed by the data-publication term of the cost function

$$C_{tx} = \frac{G_{data}}{N_{batch}} + G_{meta},$$

where G_{data} represents gas for uploading compressed transaction data and N_{batch} the batch size. Crapis et al. (2023) demonstrate that Ethereum's EIP-4844 "blob" design lowers G_{data} by approximately 90 %, translating into an 8–10× fee reduction for rollups.

From a broader economic lens, Sanka and Cheung (2021) emphasized that while off-chain batching reduces per-transaction cost, liquidity and collateral overheads can offset savings if batch utilisation is low. Lavour et al. (2022) extend cost analysis to energy and resource efficiency, arguing that zk-based batching reduces on-chain computation intensity and thereby the environmental footprint of large-scale IoE deployments.

Consequently, cost efficiency improves non-linearly with transaction volume. High-frequency applications such as exchanges or payment hubs benefit most, whereas sporadic users may find the fixed overhead of batch publication comparable to direct Layer 1 fees.

4.4 Scalability Boundaries and Hybrid

Models

Although Layer 2 solutions greatly enhance performance, they remain bound by Layer 1 bandwidth and data-availability limits. Zhou et al. (2020) note that sustained scalability depends on concurrent evolution of base-layer consensus and block-propagation protocols.

Rollups cannot exceed the data capacity of the underlying chain; thus, long-term scaling requires integrated protocol upgrades such as sharding and dank-sharding. Hardware-assisted designs are emerging to push these boundaries. Wen et al. (2025) introduce *TeeRollup*, which leverages trusted-execution environments (TEEs) to off-load proof generation and verification, cutting delay and gas by utilizing secure hardware enclaves. This hybrid cryptographic-hardware model merges software proofs with attested execution to boost throughput beyond purely mathematical limits. Economic coordination across hybrid architectures also matters. Chaliasos et al. (2024b) analyses transaction-fee mechanisms (TFMs) for zk-rollups and highlights the need for dynamic pricing that adapts to network load to prevent congestion migration from L2 to L1. Collectively, these approaches indicate that sustainable scalability arises from layering improvements combining cryptography, hardware acceleration, and adaptive economics to achieve an equilibrium between throughput, security, and cost.

4.5 User Experience and Interoperability

Ultimately, technological success depends on usability and interoperability. State Channels provide instantaneous response but require users to remain online or employ external watchers. Khalil and Gervais (2017) introduced *Revive*, a re-balancing protocol that allows off-chain liquidity adjustments without closing channels, reducing the operational friction of constant channel management. In broader networks, Burchert et al. (2018) demonstrated that channel factories multi-user smart contracts creating numerous channels in a single transaction cut setup cost by up to 90 % and improve overall routing efficiency. Privacy is another decisive dimension where Green and Miers (2017) proposed *Bolt*, an anonymous payment-channel system employing zero-knowledge commitments to conceal payment origin and amount, thereby integrating privacy with performance.

Metric	Formula /Indicator	Typical Range	Layer 2 Type with Advantage	Sources
Throughput ratio	$R = \frac{TPS_{L2}}{TPS_{L1}}$	100–300×	Rollups & Channels	Rebello 2024; Tremblay 2022
Gas cost per tx	$C_{tx} = \frac{G_{data}}{N_{batch}} + G_{meta}$	≤ 10 % of L1 cost	ZK-Rollups	Crapis 2023
Capital efficiency	$E_{cap} = \frac{V_{tx}}{(L_{locked} \cdot t_{lock})}$	High for short-term use	State Channels	Miller 2019 (adapted)
Finality latency	Time to irreversible state	< 1 min (ZK) – 7 days (OR) – < 1 s (Channels)	Varies	Qi 2024 co

Table 4: Summary of Key Metrics

These developments show that user-oriented refinements evolve alongside protocol-level scaling. Rollups now offer web-based wallets with gas-abstracted payments, while channels integrate watchtower incentives for automated safety. Seamless interoperability between these mechanisms will likely define the next generation of blockchain usability, allowing end-users to experience instantaneous, low-cost transactions without understanding the underlying Layer 2 complexity.

5 Conclusion and Future Recommendations

This research sets out to evaluate blockchain scalability at the Layer 2 level through a comparative study of Rollups and State Channels. The analysis demonstrates that both mechanisms successfully address the throughput and cost limitations of traditional Layer 1 blockchains while maintaining decentralization and security. Quantitative models confirm that Rollups improve throughput by two orders of magnitude and reduce gas costs to less than ten per cent of base-layer fees, whereas State Channels achieve instantaneous confirmation with negligible transaction cost but require continuous user participation and collateral lockups. The study reveals that each method serves a distinct functional niche. Rollups, whether optimistic or zero-knowledge, are most suitable for large-scale, multi-user environments requiring universal access, contract composability, and verifiable integrity. State Channels remain optimal for repetitive bilateral interactions such as micropayments or gaming, where latency must approach real-time levels.

Rather than competing, these technologies complement one another: Rollups expand global transaction capacity, while Channels deliver local responsiveness.

Future scalability should focus on hybrid architecture that integrates the strengths of both systems. Channels can operate within Rollup environments to provide instantaneous settlement, with Rollups anchoring cumulative results securely to Layer 1. Further research should also examine automated liquidity management, decentralized sequencer coordination, and energy-efficient proof generation to enhance sustainability. Standardized benchmarking frameworks and open data repositories will allow fair comparison across emerging implementations. As cryptographic proofs, hardware acceleration, and protocol interoperability advance, the combined Layer 2 ecosystem is positioned to deliver secure, low-latency, and cost-effective infrastructure capable of supporting Web-scale decentralized applications.

Note: This paper is a literature-based comparative study conducted as part of undergraduate research. It has not undergone peer review.

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